MATH 3603 Homework 2 Solutions

1. (i) Let h and t be the number of heads and tails obtained in the n flips, respectively. Then, $h + t = n$ and $h - t = X$. Thus $X = n - 2t$. Since t ranges from 0 to n, the possible values of X are given by the set

$$\{n - 2t \mid t \in \{0, 1, \ldots, n\}\}.$$

(ii) When $n = 3$, the possible values of X are $\{-3, -1, 1, 3\}$. We have

$$\begin{aligned}
P(X = -3) = P(h = 0, t = 3) &= \binom{3}{0}\left(\frac{1}{2}\right)^3 = \frac{1}{8},\\
P(X = -1) = P(h = 1, t = 2) &= \binom{3}{1}\left(\frac{1}{2}\right)^3 = \frac{3}{8},\\
P(X = 1) = P(h = 2, t = 1) &= \binom{3}{2}\left(\frac{1}{2}\right)^3 = \frac{3}{8},\\
P(X = 3) = P(h = 3, t = 0) &= \binom{3}{3}\left(\frac{1}{2}\right)^3 = \frac{1}{8}.
\end{aligned}$$

Thus the probability mass function of X is given by

$$p(x) = \begin{cases} \frac{1}{8}, & x = -3, 3,\\ \frac{3}{8}, & x = -1, 1,\\ 0, & \text{otherwise.} \end{cases}$$

2. (a) We must have $\int_{\mathbb{R}} f(x)\,dx = 1$. For the given f, this is equivalent to $\int_0^2 c(4x - 2x^2)\,dx = \frac{8}{3}c = 1$. Hence $c = \frac{3}{8}$.

(b) We have

$$\mathbb{P}(\frac{1}{2} < X < \frac{3}{2}) = \int_{1/2}^{3/2} \frac{3}{8}(4x - 2x^2)\,dx = \frac{11}{16}.$$

(c) We have

$$\mathbb{E}(X) = \int_0^2 x \cdot \frac{3}{8}(4x - 2x^2)\,dx = \frac{3}{8}\int_0^2 (4x^2 - 2x^3)\,dx = 1,$$

$$\mathbb{E}(X^2) = \int_0^2 x^2 \cdot \frac{3}{8}(4x - 2x^2)\,dx = \frac{3}{8}\int_0^2 (4x^3 - 2x^4)\,dx = \frac{6}{5},$$

and

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \frac{6}{5} - 1 = \frac{1}{5}.$$

3. Let $Y := \max\{X_1, \ldots, X_n\}$ and $Z := \min\{X_1, \ldots, X_n\}$, and denote their distribution functions as F_Y and F_Z, respectively. Note that $Y \le y$ if and only if $X_1 \le y, \ldots, X_n \le y$, and $Z > z$ if and only if $X_1 > z, \ldots, X_n > z$. Thus, by the independence of the X_i's, we have

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \le y) = \mathbb{P}(X_1 \le y, \ldots, X_n \le y) = \prod_{i=1}^n \mathbb{P}(X_i \le y) = F(y)^n, \\ F_Z(z) &= \mathbb{P}(Z \le z) = 1 - \mathbb{P}(Z > z) = 1 - \mathbb{P}(X_1 > z, \ldots, X_n > z) \\ &= 1 - \prod_{i=1}^n \mathbb{P}(X_i > z) = 1 - (1 - F(z))^n. \end{aligned}$$

4. For each $i = 1, \ldots, 365$, define a random variable X_i by

$$X_i = \begin{cases} 1, & \text{if someone's birthday happens to be on the } i\text{-th day of the year} \\ 0, & \text{otherwise.} \end{cases}$$

Note that $X = X_1 + \cdots + X_{365}$. For each X_i, we have $X_i = 0$ if and only if no one's birthday is on the i-th day of the year. Since the birthday of each person is independent and equally likely to be on any of the 365 days, we have

$$\mathbb{P}(X_i = 0) = \left(\frac{364}{365}\right)^{100}, \quad \mathbb{P}(X_i = 1) = 1 - \left(\frac{364}{365}\right)^{100}.$$

Thus

$$\mathbb{E}(X) = \sum_{i=1}^{365} \mathbb{E}(X_i) = \sum_{i=1}^{365} \mathbb{P}(X_i = 1) = 365\left(1 - \left(\frac{364}{365}\right)^{100}\right) \approx 87.58.$$

To calculate the variance, we first evaluate $\mathrm{Var}(X_i)$ and $\mathrm{Cov}(X_i, X_j)$ for each $1 \le i < j \le 365$. Observe that X_i^2 follows the same distribution as X_i, thus

$$\mathbb{E}(X_i^2) = \mathbb{E}(X_i) = 1 - \left(\frac{364}{365}\right)^{100},$$

and hence

$$\mathrm{Var}(X_i) = \mathbb{E}(X_i^2) - \mathbb{E}(X_i)^2 = \left(1 - \left(\frac{364}{365}\right)^{100}\right) - \left(1 - \left(\frac{364}{365}\right)^{100}\right)^2.$$

Meanwhile, for $i < j$, denote E_i and E_j as the events that nobody has a birthday on the i-th and j-th days of the year, respectively. We thus have $X_i X_j = 0$ if and only if $E_i \cup E_j$ occurs. Then by inclusion-exclusion principle, we have

$$\begin{aligned}
\mathbb{P}(X_i X_j = 0) &= \mathbb{P}(E_i \cup E_j) = \mathbb{P}(E_i) + \mathbb{P}(E_j) - \mathbb{P}(E_i \cap E_j) \\
&= 2\left(\frac{364}{365}\right)^{100} - \left(\frac{363}{365}\right)^{100}, \\
\mathbb{E}(X_i X_j) &= \mathbb{P}(X_i X_j = 1) = 1 - 2\left(\frac{364}{365}\right)^{100} + \left(\frac{363}{365}\right)^{100}.
\end{aligned}$$

It follows that

$$\begin{aligned}
\mathrm{Cov}(X_i, X_j) &= \mathbb{E}(X_i X_j) - \mathbb{E}(X_i)\mathbb{E}(X_j) \\
&= 1 - 2\left(\frac{364}{365}\right)^{100} + \left(\frac{363}{365}\right)^{100} - \left(1 - \left(\frac{364}{365}\right)^{100}\right)^2.
\end{aligned}$$

Finally, by applying the sum formula for variance, we have

$$\begin{aligned}
\mathrm{Var}(X) &= \sum_{i=1}^{365} \mathrm{Var}(X_i) + 2 \sum_{1 \le i < j \le 365} \mathrm{Cov}(X_i, X_j) \\
&= 365 \left(\left(1 - \left(\frac{364}{365}\right)^{100}\right) - \left(1 - \left(\frac{364}{365}\right)^{100}\right)^2 \right) \\
&\quad + (365 \times 364) \left(1 - 2\left(\frac{364}{365}\right)^{100} + \left(\frac{363}{365}\right)^{100} - \left(1 - \left(\frac{364}{365}\right)^{100}\right)^2 \right) \\
&= 365 \left(1 - \left(\frac{364}{365}\right)^{100}\right) - 365^2 \left(1 - \left(\frac{364}{365}\right)^{100}\right)^2 \\
&\quad + (365 \times 364) \left(1 - 2\left(\frac{364}{365}\right)^{100} + \left(\frac{363}{365}\right)^{100} \right) \\
&\approx 8.66.
\end{aligned}$$

5. (i) Let $D = \{(x, y) \in \mathbb{R}^2 : 0 \leq x^2 + y^2 \leq 1\}$ denotes the unit disk. The probability function f is zero outside D, hence

$$\int_D f(x)\, dx = \int_D C\, dx = C\pi = 1.$$

Thus $C = \frac{1}{\pi}$.

(ii) Let f_X be the marginal probability density function of X. Recall that f is zero outside D. For each fixed x, we have $(x, y) \in D$ if and only if $-1 \leq x \leq 1$ and $-\sqrt{1-x^2} \leq y \leq \sqrt{1-x^2}$. Thus $f_X(x) = 0$ if $|x| > 1$, and

$$f_X(x) = \int_{\mathbb{R}} f(x, y)\, dy = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi}\, dy = \frac{2\sqrt{1-x^2}}{\pi}, \quad |x| \leq 1.$$

In summary,

$$f_X(x) = \begin{cases} \frac{2\sqrt{1-x^2}}{\pi}, & -1 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$